

Predicting Sales from Advertising Expenditure

Statistical Data Analysis and Visualisation

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1. Introduction

Marketing analytics and business strategies require extensive research of the relationship between advertising expenditure and sales performance. The analysis of statistics, as Lang (2016) argues, is “to enable a knowledgeable reader with access to the original data to verify the reported results”. This paper presents a comprehensive statistical analysis of the advertising dataset from a published dataset on Kaggle, which contains 200 observations of sales figures and advertising budgets across three channels: TV, Radio, and Newspaper.

The problem statement for this research is: what are the most significant predictors of sales from the advertising channels and where should the budget be allocated? Conducting extensive statistical techniques is essential to maximize the potential of sales from the channels. The research methods include, exploratory data analysis, linear regression modelling, model diagnostics, hypothesis testing, principal component analysis are essential. The findings will explain the relationships within the data and offer practical recommendations for a practical budget allocation.

2. Context and Description of Data

The advertising dataset is populated with 200 observations of expenditures across four channels: TV, Radio and Newspapers. These are advertising channels which are recorded as numerical values by thousand dollars. The sales data is measured by the thousand units sold.

Following the dataset overview, the next step is generating a summary of the stat statistics using R-Studio, which is an open-source platform for programming and data visualization. Running built-in functions such as `summary()` and `sd()` reveal key metrics for mean, median, and range. The most widely spread expenditure is TV with a range of \$700 to \$296,400 and a mean of \$147,000. The Radio spending is more modest, with a mean of \$23,300 and range of \$0 to \$49,600. Newspaper budgets are moderately spread with a range: \$300 to \$114,000 and a mean of \$30,600. Sales shows a modest units sold ranging from 1,600 to 27,000 units with a mean of 15,130 units and a standard deviation of 5,280. The 1st quartile and 3rd quartile are also important measures that depict a clearer picture of data distribution than the mean; in this case, the sales fall between 11,000 and 19,050 units.

3. Exploratory Data Analysis

Exploratory Data Analysis sets the foundation prior to building statistical models. This is the initial stage of analysis where an observer can use “a range of statistical and unsupervised learning to understand patterns in data” (Poian, et.al, 2023). Examining the visualization from the histogram, TV spend is evenly distributed, while Radio and Newspaper are skewed to the left, with most budgets concentrated at the lower end of the scale.

The correlation matrix provides another key insight as part of EDA. Two key metrics that will be examined will be the Pearson correlation coefficient and p-value. The coefficient value, closest to

1, indicates strongest association and -1 as the weakest, as well as p-value under 0.05 as significant and not random. TV spending showed a strong positive association with sales with a correlation of 0.901 and p-value less than 0.001, indicating that increased spending in TV advertising leads to increased sales in roughly a linear fashion. Radio shows a moderate positive correlation of 0.350 with a p-value less than 0.001, while Newspaper shows a weaker correlation of 0.158 with a p-value of 0.026. Although the Newspaper correlation is statistically significant, it can be regarded as negligible, as it is approximately 2.5% of the variance in sales.

Figure 1. Pearson Correlation Matrix

	TV	Radio	Newspaper
TV	1	0.055	0.057
Radio	0.055	1	0.354
Newspaper	0.057	0.354	1
Sales	0.901	0.35	0.158

Pairwise scatter plots of sales against each predictor visually confirmed these findings. The Sales vs TV plot showed a tight, linear upward trend with little scatter. The Sales vs Radio plot showed a weaker but positive trend, and the Sales vs Newspaper plot showed a near-flat relationship with wide dispersion around the trend line.

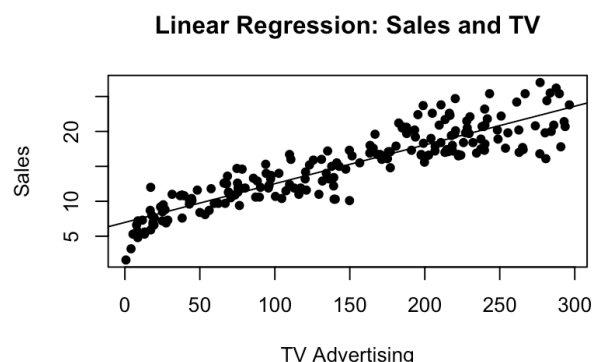
4. Regression Analysis

Regression is key to understanding how it is used to “establish causal relationship between the predictor variables and response variable” (Zhou et al., 2021). A simple linear regression model is fitted with Sales as the response variable and TV as the sole predictor.

4.1 Simple Linear Regression Model

The slope coefficient for TV advertising is 0.05, meaning that for every \$1,000 increase in TV spending, the expected sales increase is around 55 units. An intercept of 6.975 means the predicted sales would be this level if TV spending was at zero. Due to its linear progression, it can be regarded as statistically robust; however, it is not solely practical since the business could be spending in other advertising channels. The model explains 81.22% of the variance in sales ($R^2 = 0.8122$), which is a strong result for a single-predictor model and highlights the dominant influence of TV advertising on sales performance.

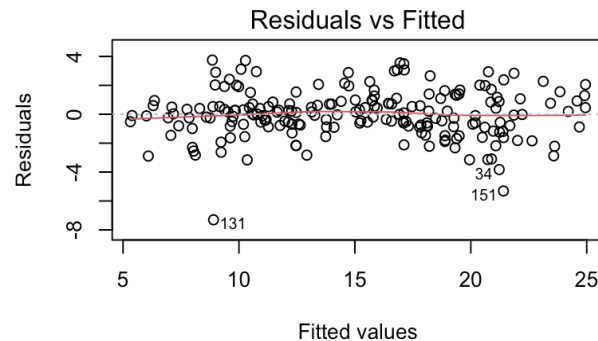
Figure 2. Simple linear regression coefficients (Sales ~ TV)



4.2 Multiple Linear Regression Model

To assess the combined effect of all three advertising channels, a multiple linear regression model was fitted with TV, Radio, and Newspaper as predictors simultaneously.

Figure 3. Multiple linear regression coefficients ($Sales \sim TV + Radio + Newspaper$)



Including all advertising channels simultaneously produces a summary report of correlation matrix with coefficients, standard errors, p-values, and R^2 / adjusted R^2 . The findings show TV and Radio are statistically significant at the 5% level as they have a p-value less than 0.05. Newspaper shows the least significant, as it has a p-value of 0.954. Radio advertising has a coefficient of 0.107 and a p-value less than 0.001, which indicates statistical significance on sales as well as the highest coefficient out of all channels. This means that for every \$1,000 spent, Radio advertisement yields around 107 units of sales. Additionally, R^2 and adjusted R^2 further suggests that there is over 90% of variance in sales.

A compelling observation can be found in Newspaper's effect between the simple and multiple regression models. With Newspaper as the sole predictor in the model, the coefficient of 0.0469 with $p = 0.026$, appears to be significant. However, in the multiple regression model, the coefficient reduces significantly to 0.0003 with p-value of 0.954. Given that Newspaper is moderately correlated with Radio (coefficient of 0.354), this could explain the effect on Sales. In simple regression, Newspaper was essentially masking Radio's effect. Once Radio is included in the model, Newspaper is shown to have a diminished effect on Sales.

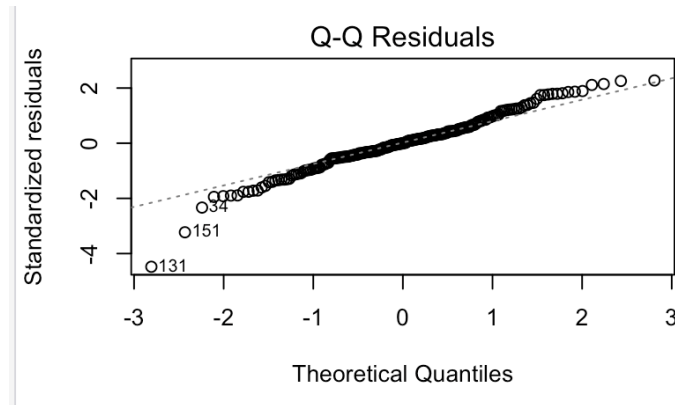
5. Model Diagnostics and Assumptions

Model diagnostics are used to test the assumptions of the multiple regression model. Firstly, in the residuals vs fitted values plot, the residuals are scattered flat near the zero-line without any clear pattern. This indicates assumptions on homoscedasticity are clear, and in other words, regarded as trustworthy. The spread of residuals was consistent across low, mid, and high fitted values with the standard deviations of 1.73, 1.37, and 1.75.

The Quantile-Quantile (Q-Q) residuals plot is also used in diagnostics to check the normality in distribution. The points standardized residuals and theoretical quantiles are nearly in a diagonal line, so it strengthens the assumption that the homoscedasticity is met. The built-in standardized

residuals function in R-studio allows various types of plots to test the assumptions, as well as displaying what the outliers are placed, which is shown as 131, 151 and 34.

Figure 4. Q-Q Residuals plot (Theoretical Quantiles versus Standardized Residuals)



6. Model Comparison and Budget Allocation

After the models are created and tested, businesses can proceed to plan for budget allocation by comparison. With a simple regression with TV, the R^2 is 0.8122, and by adding Radio and Newspaper, it goes up to 0.9026. The adjusted R^2 also goes up from 0.8112 to 0.9011. Standard residual error with TV only is 2.296 and with Radio and Newspaper then becomes 1.662, confirming that the improvement is genuine. Radio is the primary driver of this improvement and exposes Newspaper as an inadequate contributor to sales.

Figure 5. Simple vs Multiple Regression Model Comparison

Model	R^2	Adjusted R^2
Simple (TV only)	0.8122	0.8118
Multiple (TV + Radio + Newspaper)	0.9026	0.9011

For practical budget allocation, suppose a marketing manager has an additional \$10,000 to allocate to a single channel. Using the multiple regression coefficients, allocating the entire budget to TV yields 544 units sold and Radio yields 1070 units sold, a near double return. The Radio coefficient is higher than that of TV, therefore it is recommended to allocate the budget to Radio. This is carrying the assumption that all other variables are constant, the model coefficients are stable, and no interaction effects exist between TV and Radio. From a marketing manager standpoint, best business practice includes “conveying their messages or brands to their concerned audience” (Zhou, et.al,2021). Recommendation should be made invest both TV and Radio, with a heavier investment in Radio.

7. ANOVA and Alternative Testing Methods

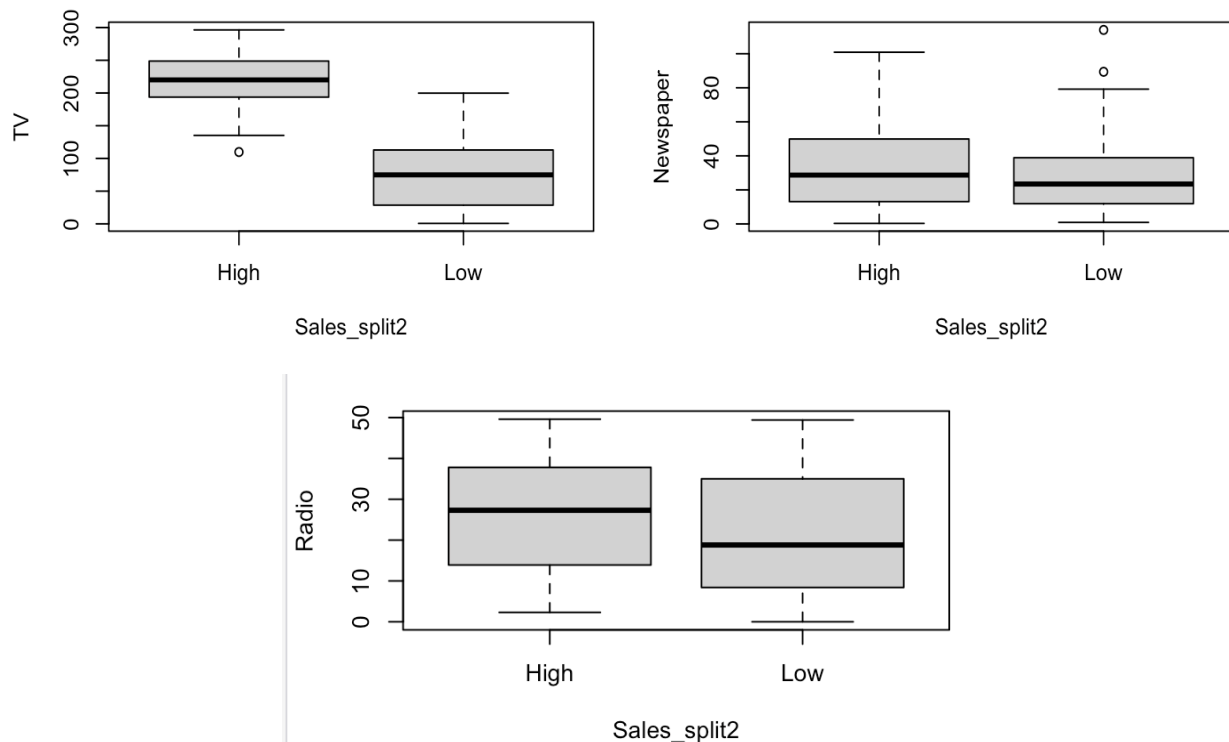
ANOVA (Analysis of Variance) is conducted to determine whether there are additional tendencies between multiple independent groups. Various testing methods such as t-test and non-parametric test are useful in this analysis to see whether each spend in channels differs systematically across markets with different sales performance.

7.1 Two-Group Split

Sales data was divided into High and Low groups at the median (sales-split2) and then running the t-test to assess differences in central tendencies of the predictor variables with sales-split2. The null hypothesis is formed to assume there is no significant difference in variables, and in this case, the mean of each individual predictor is equal between the splitted high and low sales groups. The alternative hypothesis is that the mean of each individual predictor is different between the low and high.

To visualize the variance, boxplots are used to show which channel difference is the most striking. TV displays the most distinct difference from the high and low sales group. Radio also shows a significant difference; however, Newspaper shows the weakest difference. This result assures that the Newspaper has a limited relationship with Sales.

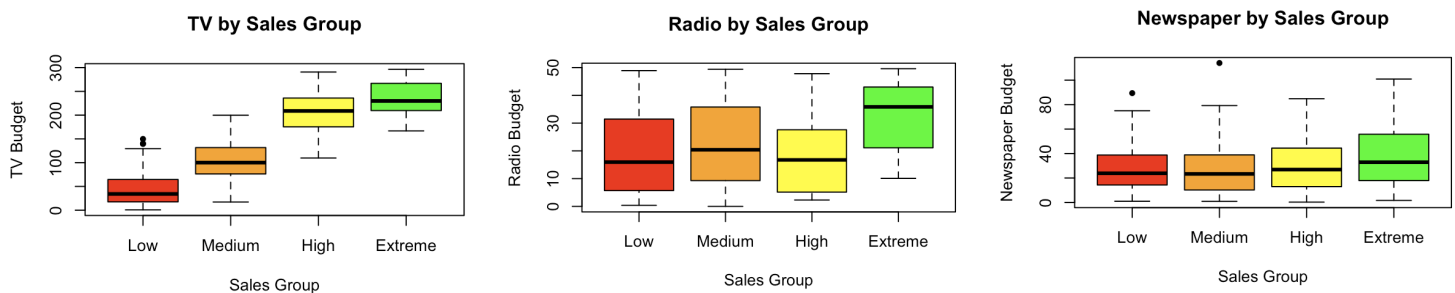
Figure 6. Four Group Split by Low and High Sales Group



7.2 Four-Group Split

The sales can be also divided into four quartiles (Low, Medium, High, Extreme), as it provides a more detailed assessment of the differences. ANOVA testing can now be applied as it is designed to compare three or more groups simultaneously. The boxplot shows TV has a steady increase from Low to Extreme, while Radio increases with more overlap in the Medium and High groups. Newspaper remains nearly flat across all four groups, which confirms the lack of relationship with sales.

Figure 7. Four Group Split by Quartiles



The non-parametric test is also applied as an alternative. Using `kruskal.test()` function derived from Kruskal-Wallis test, Newspaper with p-value is 0.3848 further suggests that there are no significant differences across the four groups.

The post-hoc test is conducted to follow up after the ANOVA tests. Across all groups, (ie. Low to High, Medium to Extreme, etc.) with adjusted p-value of less than 0.05, TV still stands out as the strong difference. In contrast, every single pair for Newspaper has an adjusted p-value over 0.05, indicating that none are significant.

8. PCA and Hierarchical Clustering

Principal Component Analysis is conducted using the predictor variables (TV, Radio, Newspaper) to identify the underlying structure of advertising spend patterns across markets.

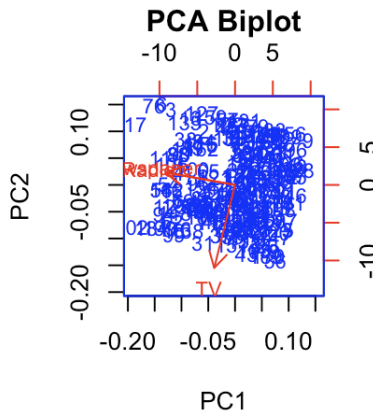
8.1 Principal Component Analysis

Three variables will be reorganised into three principal components (PC1, PC2, PC3). Under the cumulative proportion of the PCA summary, the combination of PC1 and PC2 results in 78.5% of the total variance. This provides a reasonable two-dimensional summary of the data.

The biplot is an appropriate visualization to convey the interpretation. TV arrow projects towards PC1, which is a completely different direction from the Radio and Newspaper arrows projecting towards PC2. This indicates that the TV variable is independent of the other two variables. Radio

and Newspaper arrows point approximately to the same direction, which reflect their moderate correlation as discovered in the earlier regression models.

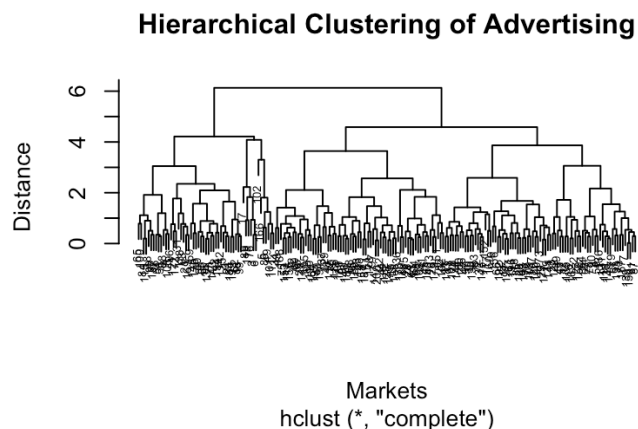
Figure 8: Principal Component Biplot



8.2 Hierarchical Clustering

Hierarchical clustering is a method to group all variables together to observe any similarities to each other. The cluster not only includes the predictor variables (TV, Radio, Newspaper) but response variables (Sales) as well.

Figure 9. Hierarchical Clustering of Advertising



The cluster dendrogram plot visualizes from bottom up the merge of all of the groups. The plot shows the larger of the two groups containing the left two thirds of the markets (x-axis) The markets merge together at lower distances (y-axis), which indicates that they are quite similar to each other. This suggests that these left two groups of markets are moderate to low spend across all advertising channels with consistent sales figures. The right group splits in a higher distance, around 6, and these markets are more diverse from each other. The diverse market in this group suggests a higher spend in advertising. Examining hierarchical clustering means that

higher TV and Radio spending consistently achieved higher sales, which was initially developed from the regression models.

9. Conclusion

The findings from the various types of models and analysis, it is statistically proven that TV and Radio are the most significant predictor of sales performance. Running multiple linear regression allowed the comparison of which channel contributes the most while the simple model concealed the accurate predictor of sales (Newspaper). An important observation was made in the correlation matrix where Radio and Newspaper had moderate correlation, which indicates that Newspaper's contribution to sales was not on its own. Methods such as model diagnostics, PCA, ANOVA and its alternative testing were conducted to solidify the findings from the regression models as well as to identify where the budget should be allocated most to. Based on the analysis it is recommended that TV and Radio should be invested together more, with Radio leading to most sales.

References

Lang, T.A. and Altman, D.G. (2016) 'Basic statistical reporting for articles published in biomedical journals: the "Statistical Analyses and Methods in the Published Literature" or SAMPL guidelines', in Science Editors' Handbook. European Association of Science Editors. Available at: <https://www.equator-network.org/wp-content/uploads/2013/07/SAMPL-Guidelines-6-27-13.pdf> (Accessed: 2 March 2026).

Poian, V.D. et al. (2023) 'Machine learning for the prediction of interplanetary magnetic field and solar wind properties', *Frontiers in Astronomy and Space Sciences*, 10. Available at: <https://www.frontiersin.org/journals/astronomy-and-space-sciences/articles/10.3389/fspas.2023.1134141/full> (Accessed: 12 March 2026).

Zhou, Z. et al. (2021) 'Impact of YouTube advertising on sales with regression analysis and statistical modeling: usefulness of online media in business', *Computational Intelligence and Neuroscience*, 2021. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8443353/> (Accessed: 18 March 2026).

Ashydv (date unknown) Advertising Dataset. Kaggle. Available at: <https://www.kaggle.com/datasets/ashydv/advertising-dataset> (Accessed: 25 February 2026).